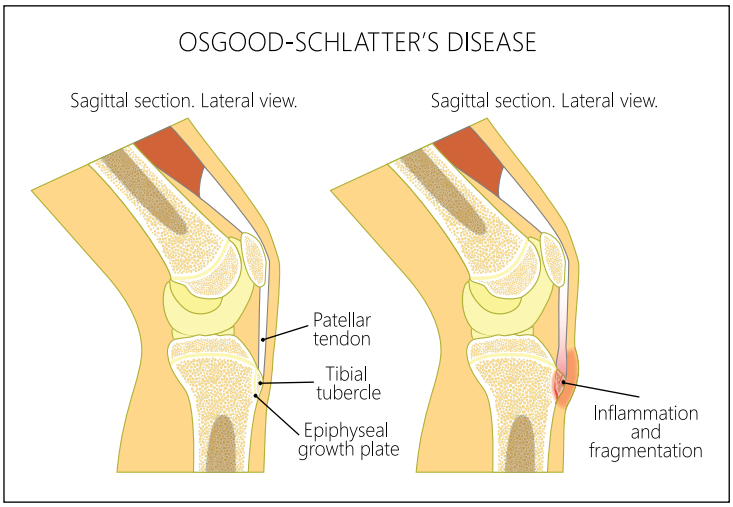
Osgood-Schlatter Disease

Background

- Osgood-Schlatter disease is an overuse injury and a common cause of knee pain in active, "skeletally immature" people under 24 years of age. "Skeletally immature" is when a person's growth plates haven't closed and they're still growing.
- It's thought that Osgood-Schlatter disease is due to repetitive overuse. The overuse leads to inflammation of and sometimes small cracks in the growing bone where the patellar (kneecap) tendon attaches to the tibia (the larger of the 2 lower leg bones). Though the bone can look fractured on imaging, the small cracks typically heal well over time as the bone matures and the growth plates close.
- People most at risk for Osgood-Schlatter disease are males aged 12–15 and females aged 8–12 It's also seen in young people who are experiencing a sudden growth spurt, those engaged in repetitive activities involving jumping and sprinting, and individuals with poor flexibility.
- Symptoms include pain and tenderness in the front of the knee, just below the kneecap, that most commonly occurs without a specific traumatic event.
- Treatment includes pain relief with ice and anti-inflammatory medications (NSAIDS such as ibuprofen or motrin), activity modification, and physical therapy.
- Osgood-Schlatter disease usually resolves on its own,, and patients typically "grow out of it" with time.

Latest Research

- Completely stopping sports activities isn't recommended since losing conditioning can lead to recurrence of Osgood-Schlatter disease or other injuries when a person returns to their sports activities. Activity modification, however, is important when experiencing symptoms.
- Patients with persistent pain after 3 months of rehabilitation may benefit from "dextrose prolotherapy" injections.





Rehab Instructions

Hamstring Stretch

1. Sit at the edge of a chair with your affected leg straight out in front of you.
2. Lean forward by bending at your hips. Be sure to keep your chest up and your back flat.
3. Bend forward until you feel a comfortable stretch in your hamstring.
4. Hold for 30–60 seconds.

Complete 2 sets 2 x a day.

Quadriceps Stretch



1. Lie down on your stomach.
2. Bend the knee of your affected leg and reach back and grasp that foot in your hand. If your hamstring is too tight, loop a towel around your ankle and pull the towel with your hand.
3. Keep your knees together and pull your heel toward your buttock until you feel a stretching sensation in the front of your thigh..
4. Hold for 30–60 seconds, then slowly lower your foot back to the start position.

Complete 2 sets 2 x a day

Sit to Stand (single leg squat with a chair)



1. Put a chair against a wall for stability.
2. Sit on the edge of the chair with your affected leg back and the leg other forward.
3. As you stand up, your affected leg will straighten first.
4. Once you're standing up, slowly lower yourself back onto the chair.
5. Keep your chest up and don't let your back arch or bend.
6. Use a chair or stool with a high seat height to make the exercise easier. Start with what's tolerable!

Complete 2 sets of 20 reps 1 x a day. Progress to 3 sets of 20 reps 1x a day.

Lateral Step Downs



1. Start with both feet on a stair or step stool and hold on to something for balance.
2. Bend the knee of your affected leg and slowly lower your other foot toward the floor. Be sure to press your hips back as if you're preparing to sit in a chair.
3. Lightly tap your heel to the floor without putting your weight on it and slowly return to the starting position.
4. The leg you're standing on is doing the work and you should feel its outer hip muscle working.
5. Don't let the knee of your affected leg collapse inward.

Complete 2 sets of 10 reps 1 x a day. Progress to 3 sets of 10 reps 1 x a day.

Practical Tips for Success

- Playing is permitted if pain level is less than a 3 out of 10 and resolves within 24 hours.
- Activities involving prolonged squatting or kneeling may cause symptoms and require changing positions.
- A continuous resting period of approximately 12 weeks (often between sports seasons) is typically required to overcome Osgood-Schlatter disease. Until this rest period is completed, persistent pain with exercise can be expected.

References

Wall, E. J. (2010). Osgood-Schlatter disease: Practical treatment for a self-limiting condition. *The Physician and Sportsmedicine, 38*(4), 29–34. doi:10.3810/psm.1998.03.802

Wu, Z., Tu, X., & Tu, Z. H. (2021). Hyperosmolar dextrose injection for Osgood–Schlatter disease: A double-blind, randomized controlled trial. *Archives of Orthopaedic and Trauma Surgery, 142*(9), 2279–2285. doi:10.1007/s00402-021-04223-1

Photos used with permission in collaboration with the Musculoskeletal Injury Rehabilitation Research for Operational Readiness (MIRROR) program and the Geneva Foundation.